Basic of Polynomials

Let S be one of the sets $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

- **Definition 1**
  A **polynomial** with coefficients in S is a formal expression

  $$p(x) = a_0 + a_1 x + \cdots + a_n x^n$$

  where $a_0, a_1, \cdots, a_n \in S$ and n is a nonnegative integer.
  The set of all polynomials with coefficients in S is denoted by $S[x]$.

- **Definition 2**
  Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n$. If $a_n \neq 0$, we say that the **degree** of $p(x)$ is n and is denoted

  $$\deg(p(x)) = n.$$

  Moreover, we say that the **leading coefficient** of $p(x)$ is a_n.
  - **Example 2.1**

    $$\begin{aligned} p(x) &= 2 + x - 3x^2 + 7x^4 \in \mathbb{Z}[x] \\ p(x) &= 1 + \frac{2}{3}x - \frac{5}{2}x^3 \in \mathbb{Q}[x] \\ p(x) &= x + \sqrt{2}x^2 - \frac{1}{7}x^5 \in \mathbb{R}[x] \\ p(x) &= \mathrm{i} - 2x + (\mathrm{i} + 1)x^2 \in \mathbb{C}[x] \end{aligned}$$

  - **Remark 2.2**
    The degree of the zero polynomial $\mathbb{0}(x) = 0$ is undefined.

- **Definition 3**
  A polynomial $p(x) \in S[x]$ is called **monic** if its leading coefficient is 1.
  - **Example 3.1**
    Let $p(x) = 1 - 5x^2 + \dfrac{\sqrt{2}}{2}x^7 + x^9 \in \mathbb{R}[x]$, then $p(x)$ is a monic polynomial with $\deg(p(x)) = 9$.

- **Definition 4**
  Two polynomials $p(x), q(x) \in S[x]$ are **equal** if $\deg(p(x)) = \deg(q(x))$ and their corresponding coefficients are equal.

- **Definition 5**
  Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n$, $q(x) = b_0 + b_1 x + \cdots + b_m x^m$. Then

  $$p(x) + q(x) = \sum_{k=0}^{\max(n,m)} (a_k + b_k)\, x^k.$$

  $$p(x)q(x) = \sum_{k=0}^{n+m} c_k x^k \quad \text{where} \quad c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

  - **Example 5.1**
    Let $p(x) = 5 + 2x + 6x^2$ and $q(x) = 3 + x + 4x^2 + 5x^3$, then

    $$\begin{aligned} &a_0 = 5, \quad a_1 = 2, \quad a_2 = 6, \quad a_3 = 0, \quad a_4 = 0, \quad a_5 = 0 \\ &b_0 = 3, \quad b_1 = 1, \quad b_2 = 4, \quad b_3 = 5, \quad b_4 = 0, \quad b_5 = 0. \end{aligned}$$

    Then

$$\begin{aligned} p(x)+q(x) &= \sum_{k=0}^{3}(a_k+b_k)\,x^k \\ &= (a_0+b_0)+(a_1+b_1)\,x+(a_2+b_2)\,x^2+(a_3+b_3)\,x^3 \\ &= (5+3)+(2+1)\,x+(6+4)\,x^2+(0+5)\,x^3 \\ &= 8+3x+10x^2+5x^3. \end{aligned}$$

$$\begin{aligned} p(x)q(x) &= \sum_{k=0}^{5} c_k x^k \quad \text{where} \quad c_k=\sum_{j=0}^{k} a_j b_{k-j} \\ &= c_0+c_1x+c_2x^2+c_3x^3+c_4x^4+c_5x^5 \end{aligned}$$

We compute each coefficient c_k directly as follows

- $c_0=a_0b_0=5\cdot 3=15.$
- $c_1=a_0b_1+a_1b_0=5\cdot 1+2\cdot 3=5+6=11.$
- $c_2=a_0b_2+a_1b_1+a_2b_0=5\cdot 4+2\cdot 1+6\cdot 3=20+2+18=40.$
- $c_3=a_0b_3+a_1b_2+a_2b_1+a_3b_0=5\cdot 5+2\cdot 4+6\cdot 1+0\cdot 3=25+8+6+0=39.$
- $c_4=a_0b_4+a_1b_3+a_2b_2+a_3b_1+a_4b_0=5\cdot 0+2\cdot 5+6\cdot 4+0\cdot 1+0\cdot 3=10+24=34.$
- $c_5=a_0b_5+a_1b_4+a_2b_3+a_3b_2+a_4b_1+a_5b_0=5\cdot 0+2\cdot 0+6\cdot 5+0+0+0=30.$

Therefore, the product polynomial is

$$p(x)q(x)=15+11x+40x^2+39x^3+34x^4+30x^5.$$

- The arithmetic properties of $S[x]$ are as follows
  1. $$(p(x)+q(x))+r(x)=p(x)+(q(x)+r(x)).$$
  2. $$p(x)+\mathbb{0}(x)=p(x).$$
  3. For each $p(x)\in S[x]$, there exists $-p(x)\in S[x]$ such that $p(x)+(-p(x))=\mathbb{0}(x)$.
  4. $$p(x)+q(x)=q(x)+p(x).$$
  5. $$(p(x)q(x))r(x)=p(x)(q(x)r(x)).$$
  6. $$p(x)\cdot 1=p(x).$$
  7. $$p(x)q(x)=q(x)p(x).$$
  8. $$p(x)(q(x)+r(x))=p(x)q(x)+p(x)r(x).$$

  - **Remark 5.2**
    Let $p(x),q(x)\in S[x]$ be nonzero.
    1. If $p(x)+q(x)\neq \mathbb{0}(x)$, then
    $$\deg(p(x)+q(x))\leqslant \max\left\{\deg(p(x)),\deg(q(x))\right\}.$$
    2. $$\deg(p(x)q(x))=\deg(p(x))+\deg(q(x)).$$
  - **Question 5.3**
    Can the inequality in $\mathrm{Remark}\ 5.2(1)$ be strict even when both polynomials are nonzero?
    - **Solution 5.3.1**
      Yes. Note that for $p(x)=1+x-x^2$ and $q(x)=2x+x^2$, we have $p(x)+q(x)=1+3x$. Here, $\deg(p(x)+q(x))=1<\max\{2,2\}=2$.

We use $\mathbb{K}$ to denote $\mathbb{Q},\mathbb{R}$, or $\mathbb{C}$.

- **Remark 5.4**
  Let $p(x),q(x)\in \mathbb{K}[x]$. If $p(x)q(x)=0$, then $p(x)=0\vee q(x)=0$. In fact, suppose that $p(x)q(x)=0$ and $p(x)\neq 0\wedge q(x)\neq 0$. Hence we obtain that

$$\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x)).$$

But since $p(x)q(x) = 0$, the degree of the left side is undefined, which yields a contradiction.

- **Theorem 5.5 (The division algorithm for $\mathbb{K}[x]$)**
  Let $p(x), s(x) \in \mathbb{K}[x]$ with $s(x) \neq 0$, then there exist unique polynomials $q(x), r(x) \in \mathbb{K}[x]$ such that

  $$p(x) = s(x)q(x) + r(x)$$

  with either $r(x) = 0$ or $\deg(r(x)) < \deg(s(x))$.

  - **Proof 5.5.1**
    Existence
    If $p(x) = 0$, then we take $q(x) = r(x) = 0$.
    If $p(x) \neq 0$, we use induction on $\deg(p(x))$.

    - **Base step** Suppose that $\deg(p(x)) = 0$. Since $s(x) \neq 0$, we have the following cases

      - If $\deg(s(x)) = 0$, then $s(x) = b \neq 0$, so $b^{-1} \in \mathbb{K}$. We take $q(x) = b^{-1}p(x)$ and $r(x) = 0$.
      - If $\deg(s(x)) > 0$, we take $q(x) = 0$ and $r(x) = p(x)$, where $\deg(r(x)) = 0 < \deg(s(x))$.
        Thus, the base step holds.

    - **Inductive step** Suppose that the result is true for any polynomial with degree less than or equal to $n - 1$. Now we need to check if the result is true for $\deg(p(x)) = n$. Let $\deg(p(x)) = n$ and $\deg(s(x)) = m$. We consider two cases

      - If $n < m$, we take $q(x) = 0$ and $r(x) = p(x)$.
      - If $n \geqslant m$, let $p(x) = a_0 + a_1x + \cdots + a_nx^n$ and $s(x) = b_0 + b_1x + \cdots + b_mx^m$ with $a_n, b_m \neq 0$. We construct the difference

        $$\begin{aligned} p(x) - b_m^{-1}a_nx^{n-m}s(x) &= (a_0 + a_1x + \cdots + a_nx^n) \\ &\quad - b_m^{-1}a_nx^{n-m}\left(b_0 + b_1x + \cdots + b_mx^m\right) \\ &= (a_0 + a_1x + \cdots + a_nx^n) \\ &\quad - \left(b_m^{-1}a_nb_0x^{n-m} + \cdots + a_nx^n\right). \end{aligned}$$

        Notice that the leading term a_nx^n cancels out completely. That is, $\deg\left(p(x) - b_m^{-1}a_nx^{n-m}s(x)\right) \leqslant n - 1$. By the inductive hypothesis, there exist polynomials $q'(x), r(x) \in \mathbb{K}[x]$ such that

        $$p(x) - b_m^{-1}a_nx^{n-m}s(x) = s(x)q'(x) + r(x)$$

        with $r(x) = 0$ or $\deg(r(x)) < \deg(s(x))$. Now, if we take $q(x) = q'(x) + b_m^{-1}a_nx^{n-m}$, we have

        $$p(x) = s(x)q(x) + r(x)$$

        with $r(x) = 0$ or $\deg(r(x)) < \deg(s(x))$. This completes the inductive step.

    Uniqueness
    Suppose that $p(x) = s(x)q(x) + r(x)$ with $r(x) = 0$ or $\deg(r(x)) < \deg(s(x))$, and $p(x) = s(x)q'(x) + r'(x)$ with $r'(x) = 0$ or $\deg(r'(x)) < \deg(s(x))$. Hence

    $$\begin{aligned} s(x)q(x) + r(x) &= s(x)q'(x) + r'(x) \\ s(x)(q(x) - q'(x)) &= r'(x) - r(x). \end{aligned}$$

    If $q(x) \neq q'(x)$, then since $s(x) \neq 0$, the degree of the left side is

    $$\deg(s(x)(q(x) - q'(x))) = \deg(s(x)) + \deg(q(x) - q'(x)) \geqslant \deg(s(x)).$$

    However, for the right side, since $\deg(r(x)) < \deg(s(x))$ and $\deg(r'(x)) < \deg(s(x))$, we have

$$\deg(r'(x) - r(x)) \leqslant \max\{\deg(r'(x)), \deg(r(x))\} < \deg(s(x)),$$

which yields a contradiction. Therefore, $q(x) = q'(x)$ and thus $r(x) = r'(x)$.

- **Example 5.6**
  If $p(x) = 8x^4 - 4x^3 + 2x^2 + x + 1$ and $s(x) = 2x^2 + 3x + 7$, then

$$\begin{aligned} p(x) - 4x^2 s(x) &= -16x^3 - 26x^2 + x + 1 \\ -16x^3 - 26x^2 + x + 1 + 8x s(x) &= -2x^2 + 57x + 1 \\ -2x^2 + 57x + 1 + s(x) &= 60x + 8. \end{aligned}$$

  Hence, $p(x) = s(x)(4x^2 - 8x - 1) + (60x + 8)$.

- **Definition 6**
  Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{K}[x]$. If $\alpha \in \mathbb{K}$, the **evaluation** of $p(x)$ at α is

$$p(\alpha) = a_0 + a_1 \alpha + \cdots + a_n \alpha^n,$$

  that is, $p(\alpha) \in \mathbb{K}$.

  - **Theorem 6.1 (Remainder theorem)**
    Let $p(x) \in \mathbb{K}[x]$ and let $\alpha \in \mathbb{K}$, then there exists $q(x) \in \mathbb{K}[x]$ such that

$$p(x) = (x - \alpha) q(x) + p(\alpha).$$

    - **Proof 6.1.1**
      By the division algorithm, there exist unique polynomials $q(x), r(x)$ such that

$$p(x) = (x - \alpha) q(x) + r(x)$$

      with $r(x) = 0$ or $\deg(r(x)) < \deg(x - \alpha) = 1$. This implies that $r(x)$ must be a constant polynomial, so we can write $r(x) = c$ for some $c \in \mathbb{K}$.
      We now evaluate the polynomial equation $p(x) = (x - \alpha) q(x) + c$ at the point $x = \alpha$

$$p(\alpha) = (\alpha - \alpha) q(\alpha) + c = 0 \cdot q(\alpha) + c = c.$$

      Since $c = p(\alpha)$, substituting this back into the division equation yields

$$p(x) = (x - \alpha) q(x) + p(\alpha).$$

      Therefore, the theorem is proved.